

AUGUSTINAS SIRIUS

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About Me

Software Engineering student with production experience in high throughput backend and distributed systems. Experienced in Java, C++, Python, PostgreSQL, Kafka, Docker, and numerical optimisation, with a focus on high performance and systems engineering across financial, aerospace, embedded, and robotic applications.

EXPERIENCE

Software Engineering Intern — SB Technologies

January 2026 - June 2026

- Developed and maintained **high throughput Java backend services** supporting production transaction processing workloads of **2,000+ transactions per minute**.
- Designed and implemented **RESTful APIs** and client side features using **TypeScript, JavaScript, and CSS**.
- Built and optimised **PostgreSQL data processing pipelines**, reducing processing latency from 10 ms to 6 ms (40% reduction).
- Developed **Python automation and testing tools** for machine learning validation, improving engineering workflow efficiency and reducing manual testing effort.
- Containerised applications using **Docker** and contributed to **CI/CD pipelines, deployment automation, and release processes**.

IT Intern — Arnotts

Feb 2025 – July 2025

- Provided technical support across hardware, software, operating system, application, and user access issues in an enterprise environment.
- Managed **user accounts and access controls**, including account creation, modification, deactivation, password resets, and onboarding/offboarding.
- Diagnosed networking and connectivity issues involving **DNS, DHCP, VPNs, routing, and endpoint connectivity**.
- Assisted with **endpoint security, vulnerability assessments, malware scanning, and security incident investigation and escalation**.

Digital & Direct Customer Representative — Permanent TSB

July 2022 – Present

- Delivered high volume customer support (**100+ interactions daily**) across secure financial systems.
 - Investigated and resolved disputes, technical issues, and **security related incidents** including authentication failures and fraud indicators.
 - Handled account security processes (password resets, 2FA issues, fraud prevention checks).
 - Escalated complex cases while maintaining SLA compliance in a regulated environment.
 - Achieved high first contact resolution(95%) while maintaining strict compliance with security protocols.
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EDUCATION

BSc Computer Science & Software Engineering — Maynooth University (2022 - 2027)

Relevant Coursework

Algorithms & Data Structures, Computer Architecture, Operating Systems, Software Engineering, Theory of Computation, Computer Networks, Parallel & Distributed Systems, Computer Vision, Linear Algebra, Differential Equations, Statistics, Discrete Mathematics, Numerical Computation, Mathematical Algorithm & System Analysis

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, TypeScript, Bash

Backend & Distributed Systems: REST APIs, FastAPI, Kafka, Redis, Docker, Kubernetes, Distributed Systems, Event Driven Systems

Data & Cloud: PostgreSQL, MySQL, MongoDB, AWS, GCP, ETL, CI/CD, GitHub Actions, Jenkins

Systems & Networking: Linux, Windows, TCP/IP, DNS, HTTP/S, VPNs, Firewalls, Active Directory, Entra ID

Performance & Quantitative Computing: Numerical Optimisation, C++/OpenMP, Heston Model, Parallel Computing, Prometheus, Grafana

PROJECTS

Real-Time Market Data Pipeline (Distributed Systems Engineering)

- Designed and implemented a **low latency distributed market data pipeline** using Apache Kafka, Redis, FastAPI, and Docker.
- Processed **thousands of simulated market events per second**, implementing data validation, fault handling, and resilient service communication.
- Built Prometheus/Grafana monitoring for throughput, latency, service health, and anomalies.

Heston Calibration Engine (High-Performance C++ System)

- Developed a high performance C++ calibration engine for the Heston stochastic volatility model using numerical optimisation techniques.
- Implemented OpenMP based parallel computation to accelerate computationally intensive calibration workloads.
- Applied numerical methods, parallel computing, memory aware implementation, and performance optimisation to a quantitative finance problem.

Network Anomaly Detection System (Security + Machine Learning)

- Built a hybrid machine learning pipeline for network anomaly detection using supervised Random Forest and unsupervised Isolation Forest models.
 - Implemented real time inference and alert logging using engineered network traffic features and F_2 evaluation.
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LANGUAGES : English • Lithuanian • French

CERTIFICATIONS : CompTIA PenTest+ ISC2 CC Google Cloud Digital Leader Stanford University: Machine Learning